

Ravleen Bajaj

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Education

Simon Fraser University

Sept 2024 – Present

MS in Statistics

- **Supervisor:** Dr. David Stenning
- **Coursework:** Machine Learning, Statistical Theory, Statistical Computing, Measure Theory, GLMs

University of Delhi — BS in Statistics

July 2021 – July 2024

Skills / Technologies

Programming: Python, R, Julia, SQL **Frameworks:** PyMC, NumPyro, JAX **Tools:** Git, Slurm, L^AT_EX

Projects

Emulation and Calibration using Gaussian and Deep Gaussian Processes (MSc Thesis)

2024 - Present

- Bayesian computer model emulation and calibration using (deep) Gaussian Processes (GPs).
- We examine tradeoffs between modular and non-modular approaches when doing calibration (i.e., inference).
- To efficiently incorporate the high-dimensional and non-stationary simulator response surface, we fit a DeepGP emulator on simulated stellar evolution data. We use Vecchia approximation for faster computation.
- We adopt a multi-stage, modular approach to both emulation & calibration, for pragmatic inference while retaining uncertainty quantification. We validate MCMC convergence using the Gelman-Rubin statistic.

Annealed Sequential Monte Carlo (SMC) for Breast Cancer Detection

2025

- Implemented annealed SMC model for Bayesian logistic regression with 96% accuracy.
- Applied the method to the Wisconsin Breast Cancer dataset to classify malignant vs. benign tumours.
- Estimated and visualized posterior distributions of model coefficients and credible intervals.
- Analyzed feature importance, algorithm diagnostics (ESS, MCMC acceptance rates) to assess performance.

Reduction of Small-Sample Bias of GLM Parameter Estimates

2025

- Investigated the problem of small-sample bias in maximum likelihood estimation (MLE) of GLM parameters. Implemented and compared three bias-reduction techniques: asymptotic bias correction, Firth's penalized likelihood method, and Bayesian-inspired log-F priors.
- Conducted simulation studies in R to evaluate estimator bias, efficiency, and coverage probabilities across varying sample sizes and model scenarios. Demonstrated that Firth's method and log-F priors offer improved stability and reduced bias in small-sample settings relative to classical MLE.

Experience

Research Assistant

Simon Fraser University

Burnaby, BC

Jan 2026 – Present

- Provide statistical consulting services to clients by designing experiments, implementing machine learning algorithms, developing statistical & computational models, and creating reproducible workflows in R/Python.

Graduate Teaching Assistant

Simon Fraser University

Burnaby, BC

Sep 2024 – Present

- **STAT 350:** Weekly **lecture-style** tutorials. TA in the STAT workshop, a drop-in facility for students.
- Resolving queries, helping students build concepts along with discussing issues with marking and understanding exam solutions. Marking assignments, quizzes, exams and invigilating tests.

Part-Time Tutor

Rootmaths Learning Center, Freelance

Coquitlam, BC

Feb 2025 – Jul 2026

- Tutor for undergraduate-level statistics, algebra and calculus courses. One-on-one and group tutoring for high school and middle school students for statistics, mathematics and pre-calculus.

Analyst Intern

Perfora

New Delhi, IN
Aug 2023 – Oct 2023

- Created a web scraping model to regularly record product placement on website and give suggestive product placements to boost and monitor product-wise sales and performance of individual products and combos.
- Regularly monitored inventory and failed orders using Power BI and Excel to identify any flaws or errors.
- Assessed sales data per day-to-day goals and requirements including new product launches and special sales.

Data Science Intern

DMI Finance

New Delhi, IN
June 2022

- Prepared end-to-end model reports for Data Science team and documentation of model interpretation for stakeholders. Restructured model code and mapped model documentation for efficiency.
- Interpreted and comprehended several ML models based on algorithms like XGBoost and logistic regression.

Awards

2025 Graduate Fellowship (\$7,000 CAD)	2026 Graduate Fellowship (\$7,000 CAD)
2026 Graduate Fellowship (\$3,500 CAD)	2026 Travel and Research Award (\$600 CAD)
2026 CSSC Travel Award (\$150 CAD)	

Talks and Posters

Canadian Statistics Student Conference (CSSC) <i>Emulation and Bayesian Calibration of Stellar Evolution Models Using Deep Gaussian Processes</i> 🔗	2026
Canadian Celebration of Women in Computing West (CAN-CWiC West) <i>Introduction to Gaussian Processes for Emulation and Calibration</i> 🔗	2025

Organizing & Scientific Committee

President: Department of Statistics Graduate Caucus	2025-Present
Co-Organizer: 2025-2026 SFU-UBC Joint Statistics Seminar	2025
Co-Organizer: 2025-2026 SFU-UBC Joint Statistics Seminar	2026

Volunteering

Management Lead Google Developer Student Club	New Delhi, IN 2021-2022
◦ Optimized events around the year and worked on reports, events, sessions, competitions and hackathons. ◦ Delivered “Data Science 101”, a talk designed to familiarize community members with basics using Python.	
Changemaker Program Volunteer, Analytics Volunteer Pratisandhi	New Delhi, IN Sep 2021-June 2022
◦ Worked on analysis of surveys and drew relevant inferences. Curated reports for pre and post survey analysis. ◦ Cleaned data, performed EDA and conducted analysis on surveys conducted internally and on-ground.	
Eco Commander Paryavaran Sanrakshan Gatividhi (Govt. of India Initiative)	New Delhi, IN Oct 2021
◦ Achieved “platinum” certification where I educated 100+ individuals about sustainable homes and lifestyle. ◦ Adopted sustainable practices and educated community members about creative sustainable measures.	

Certifications

Institution	Course	Status/Grade
HarvardX	Data Science: R Basics	Grade: 100%
HarvardX	Data Science: Visualisations	Grade: 100%
University of Delhi	Data Science Using Python (Certification Course)	Grade: A+
